REQUIREMENT ANALYSIS

E-commerce Website Testing – SauceDemo

Project	SauceDemo E-commerce QA Portfolio
Application	https://www.saucedemo.com/
Document type	Application overview and functional requirements
Prepared by	QA Tester
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1. Purpose

This document records the application context and the functional areas selected for the SauceDemo manual QA portfolio project. It provides a reference for designing test scenarios and test cases. It is based on the application features included in the current project scope; it is not a formal specification supplied by the application owner.

2. Application Overview

SauceDemo is a web-based demonstration shopping application used here as the system under test for practising manual software testing. The project examines key customer-facing flows, including signing in, viewing products, sorting the product list, opening product details and managing items in the shopping cart.

3. Stakeholders and User

For this learning project, the primary user role is a shopper who accesses the application through a web browser. The QA tester prepares and executes tests, records observed behaviour and documents potential defects. No actual business stakeholder or development-team contact has been specified for this portfolio exercise.

4. Functional Requirements

ID	Functional area	Requirement statement
FR-LOGIN-01	Login	The application should allow a user with valid demonstration credentials to sign in and reach the product listing page.
FR-LOGIN-02	Login	The application should validate missing username or password input and display an appropriate validation message.
FR-LOGIN-03	Login	The application should reject invalid credential combinations and display an authentication error.
FR-LOGIN-04	Login	The application should prevent the documented locked-out demonstration user from signing in and show the corresponding message.
FR-LOGIN-05	Login	The password input should mask entered characters.
FR-LOGIN-06	Login	The login form should support submission using the Enter key.

ID	Functional area	Requirement statement
FR-PROD-01	Product listing	The product listing page should display the products available in the demonstration catalogue.
FR-PROD-02	Product listing	Each listed product should show its name, description, price and product image.
FR-PROD-03	Product listing	Each product not already in the cart should provide an Add to Cart action.
FR-SORT-01	Sorting	The product list should support sorting by product name in ascending and descending alphabetical order.
FR-SORT-02	Sorting	The product list should support sorting by price from low to high and high to low.
FR-NAV-01	Navigation	Selecting a product should open the corresponding product details page.
FR-NAV-02	Navigation	The user should be able to return from product details to the product listing page.
FR-NAV-03	Navigation	The shopping cart icon should open the cart page.
FR-CART-01	Shopping cart	The user should be able to add a product from the listing page to the shopping cart.
FR-CART-02	Shopping cart	After adding a product, the interface should reflect the cart state, including the action button and cart item count where displayed.
FR-CART-03	Shopping cart	The user should be able to remove a previously added product from the listing page.

5. Business Rules and Observations to Verify

The following are testable expectations adopted for this portfolio. They should be checked against the behaviour actually observed in the application; where the application behaves differently, record the evidence rather than assuming the requirement is satisfied.

• A successful login should lead to the product listing page.

• A validation or authentication message should explain why a login attempt was not accepted.

• Sorting should rearrange the displayed products according to the selected option.

• Adding or removing a product should update the visible cart state consistently.

• Product information shown on the listing and details pages should refer to the same selected product.

6. Non-Functional Considerations

This initial project focuses on functional manual testing. Basic usability observations, such as whether labels and product information are readable and whether images appear, may be recorded during execution. Formal performance, security, accessibility and cross-browser compatibility testing are not included in the current scope.

7. Assumptions and Constraints

• The SauceDemo website is available during the planned test sessions.

• The demonstration accounts and catalogue data remain accessible.

• Testing is performed through a desktop web browser.

• Test results reflect the application behaviour observed on the date of execution and may change if the application is updated.

• This document is a portfolio-level analysis, not an official product requirements specification.

8. Traceability to Test Cases

The functional requirements are covered by the test case groups maintained in the project spreadsheet: Login (TC_LOGIN), Product Visibility and Information (TC_PROD_001–006), Product Sorting (TC_PROD_007–011), Product Navigation (TC_PROD_012–015) and Add to Cart (TC_PROD_016–018). Update this mapping if test cases are added, removed or changed.

9. Review Note

This document should be reviewed whenever the project scope or observed application features change. Any requirement that cannot be verified through the application should be marked as not verified rather than treated as confirmed.